



mendix

AI Experience



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1 Goal of this document

In this walkthrough you will get acquainted with the following Mendix components:

- AI Bot Starter app
- Conversational UI
- Token Monitoring
- Mendix GenAI Connector & GenAI Commons
- Mendix Cloud GenAI Resource Packs: Model & Embedding



2 Main prerequisites

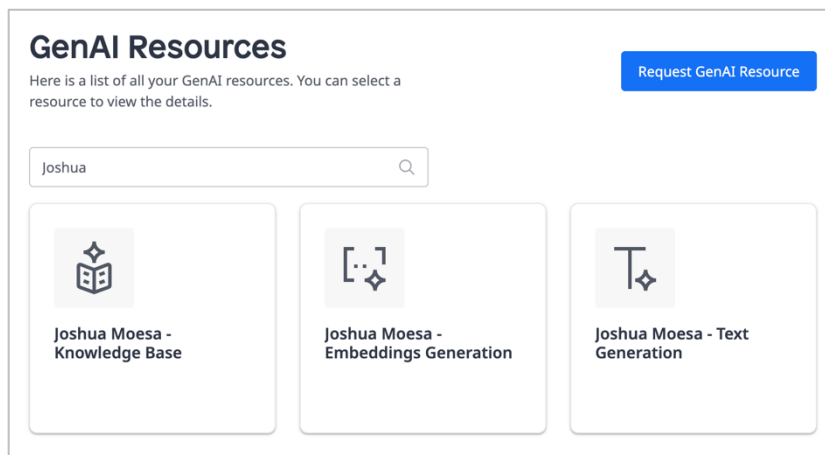
2.1 Mendix Studio Pro

This walkthrough is written for Mendix Studio Pro version **11.14**.

2.2 Mendix GenAI Resources

In your web browser, go to: <https://genai.home.mendix.com/>. Login with your Mendix account if haven't done that already.

Validate that you have Mendix GenAI resources available in the Mendix GenAI portal. You should see resources *Text generation*, *Knowledge Base* and *Embeddings Generation*

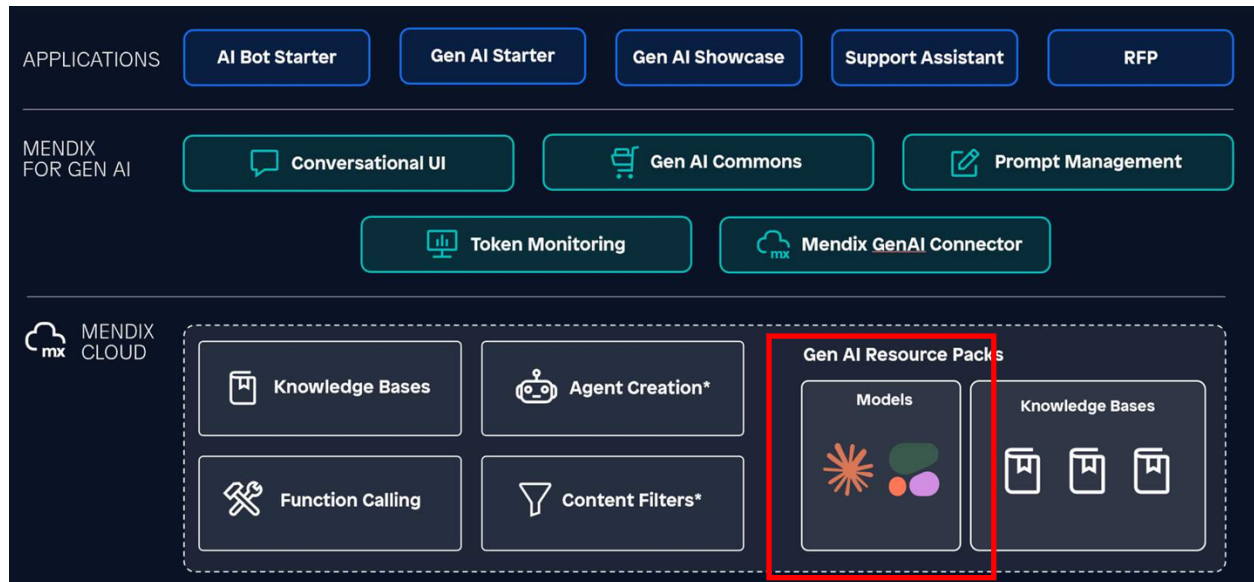


The Mendix GenAI portal is the central hub for configuring and managing Mendix GenAI resources. Target customers are customers open to Mendix Cloud, who do not yet have existing AI infrastructure that they would like to tie into.

Visit the Mendix Docs for more info:

<https://docs.mendix.com/appstore/modules/genai/mx-cloud-genai/resource-packs/>

Great job! You have finished all the prerequisites to gain access to Text Generation: the Mendix Cloud GenAI Resource Pack. This resource pack hosts the Large Language Model which we will refer to “model” from now on.



3 Exercise – AI Chatbot

In this exercise you are going to realize your own chatbot driven by AI.



The AI Bot Starter App (available to all Mendix customers in the Marketplace) serves as a starting point for developers to explore and understand Gen AI capabilities. It works with Mendix Gen AI Resources, Amazon Bedrock, OpenAI, and custom Gen AI connectors. It is also fully customizable to include advanced AI features like multimodal inputs, function calling, and RAG.

For more details, visit the Mendix documentation:

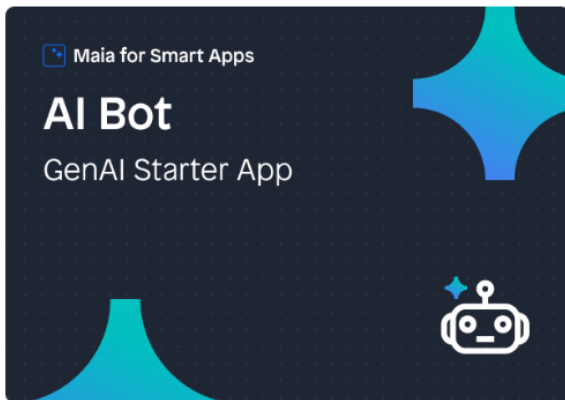
<https://docs.mendix.com/appstore/modules/genai/how-to/starter-template/>

A full demo of the AI Chatbot hosted by Mendix: <https://ama.mendixlabs.com/index.html>

3.1 Prerequisite

Validate that you have access to AI Bot Starter App. This manual is written for version **4.6.0**.

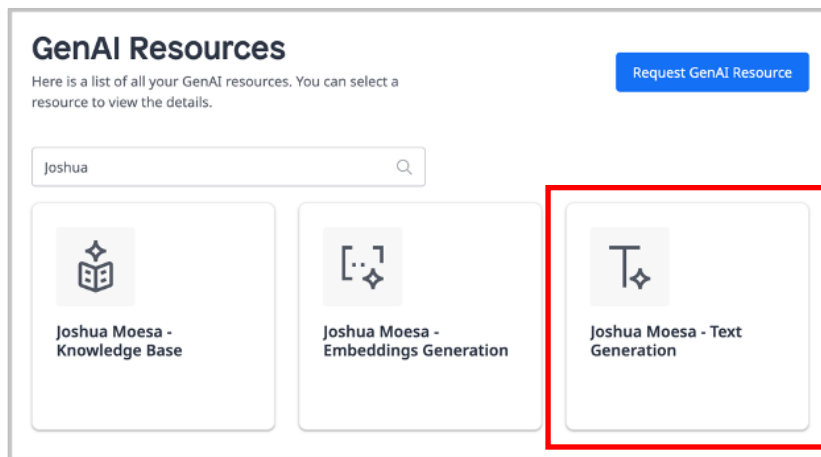
Go to: <https://marketplace.mendix.com/link/component/227926>



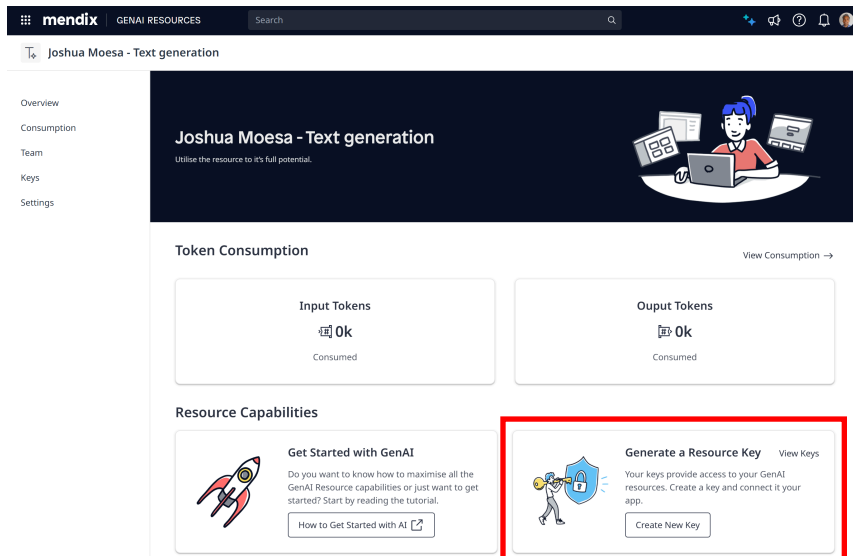
3.2 Arrange access to the GenAI resource

You are going to prepare your model resource for interaction with the AI Chatbot app. With the new API key that you are going to create, you provide your app with programmatic access.

1. Go to the Mendix GenAI portal <https://genai.home.mendix.com>
2. Click on the *Text generation* tile.



3. Click **Create New Key** button to create a new key.



4. Insert a description for your key like: "<YOUR NAME> - AI Bot" and click button **Generate API Key**.

5. Copy the created code by pressing the **Copy** button and save it in a text file on your computer for later use.



Mendix Cloud Model Resource Packs offer customers access to large language model capacity. Instantly set up GenAI environments without hyperscaler complexity. Each resource pack includes an allocation of input/output tokens for Anthropic's Claude and Cohere's Embed. Support for additional models will be introduced in the future.

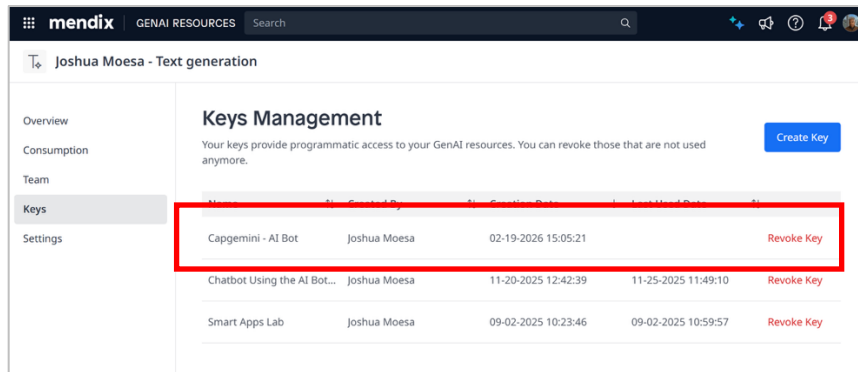
Key points:

- Choose from small, medium, or large resource pack options;
- You can revoke keys that are not used anymore;
- An activity trail provides transparency about which user has created keys.

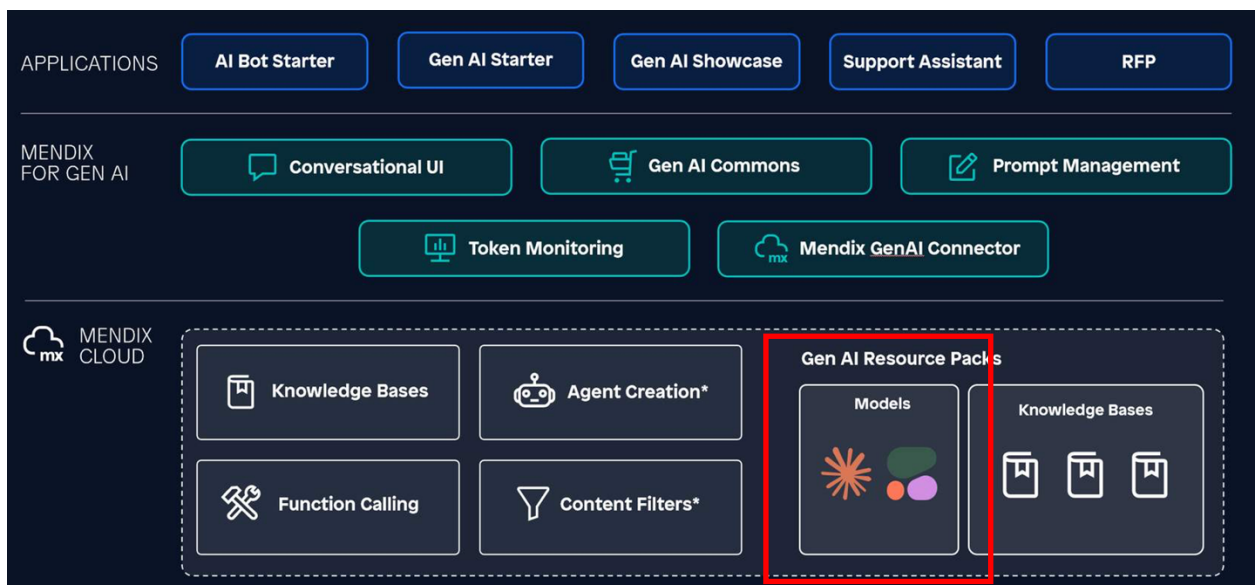
Visit Mendix Docs for more info:

<https://docs.mendix.com/appstore/modules/genai/mx-cloud-genai/resource-packs/>

The result should look similar like this:



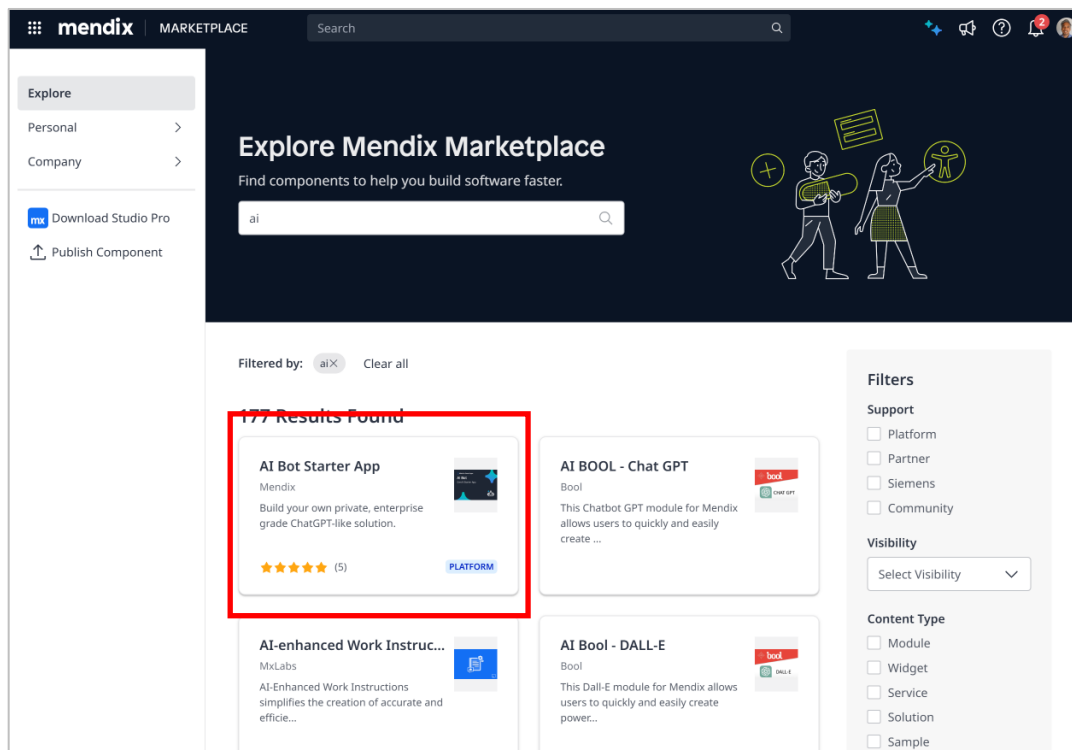
Excellent! After finishing all the prerequisites to gain access to the model, you have created an API Key to gain programmatic access using the Mendix application you are going to implement in the next section.



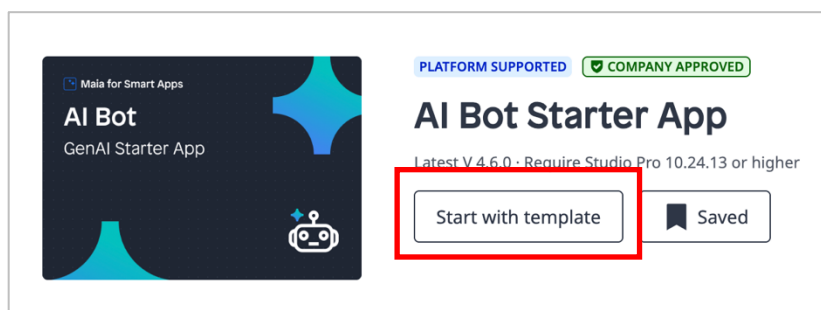
3.3 Create an AI Chatbot app from the AI Bot Starter App

1. Go to the Mendix Marketplace <https://marketplace.mendix.com> and in the search field type “ai”.
2. In the search results, press tile *AI Bot Starter App*.

If necessary: to limit the amount of search results, press filter *Platform* under the Support category.



3. On the AI Bot Starter App page press button **Start with template**.



4. In the *App Information* tab, fill in your details and press **Next**.

5. In the *Select Starter App* tab, notice that *AI Bot Starter App* is preselected. Now press button **Create App**.
6. You will be automatically redirected to the project space. Press **Edit in Studio Pro** to pull the code to your workstation.
7. In pop-up *Mendix Version Selector* choose to open the project with version **11.12.0**. If asked, convert the project.
8. The project is now opened in Mendix Studio Pro!

3.4 Configure your AI Chatbot app

To use the AI Chatbot app on your workstation, you need to complete the configuration in the Studio Pro project AND eventually in the local running app.

3.4.1 Fix the *EncryptionKey* project error in Studio Pro

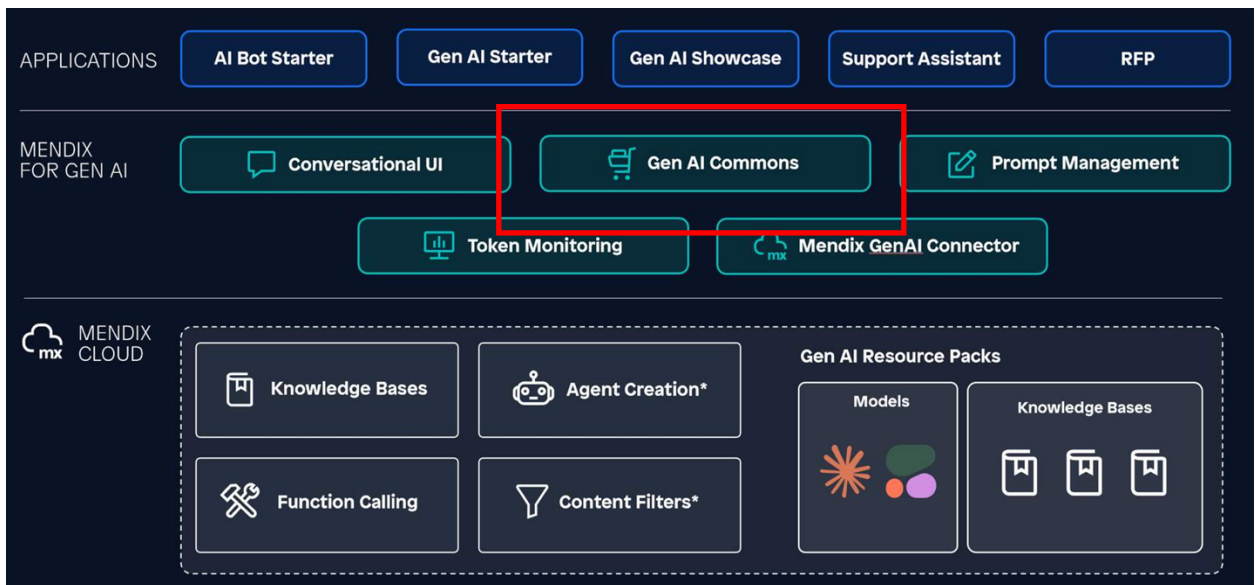
1. In Studio Pro you will notice a project error in the *Errors* tab. Double-click on the error: *The active configuration contains...Encryption.EncryptionKey*.
2. In *App Settings* pop-up double-click on the **Default** configuration row item.
3. In *Edit Configuration* pop-up click on tab **Constants**.
4. Double-click on the constant *Encryption.EncryptionKey* and in the *Edit Constant Value* pop-up in field *Value* enter a 32-character string. For example: `h[<Zrwj;LbGFL_%d1{sQy+>S@T'b3y[C`
5. Validate that the error is gone now.

3.4.2 Complete Mendix Cloud GenAI Configuration in the app

Connect your app to the Mendix Cloud GenAI portal.

1. In Studio Pro, click the green play button to run the project locally. And view the chatbot app in your web browser.
2. Log in with username `MxAdmin` and password `1`
3. After logging in, in the top bar under *Administration* click on **Mendix Cloud GenAI Configuration**
4. You are going to use the key that has been generated in step *3.2 Arrange access to the GenAI resource*. In tab *Deployed models*, press button **Import key**.
5. A pop-up appears, paste the key in the text area field and press **Import key**.
6. Test the connection with **Test key**.

You are on fire! You have successfully wired your AI Bot App to the model leveraging *GenAI Commons* components like the *Mx Cloud GenAI Connector*.



3.4.3 Complete Bot Configuration in the app

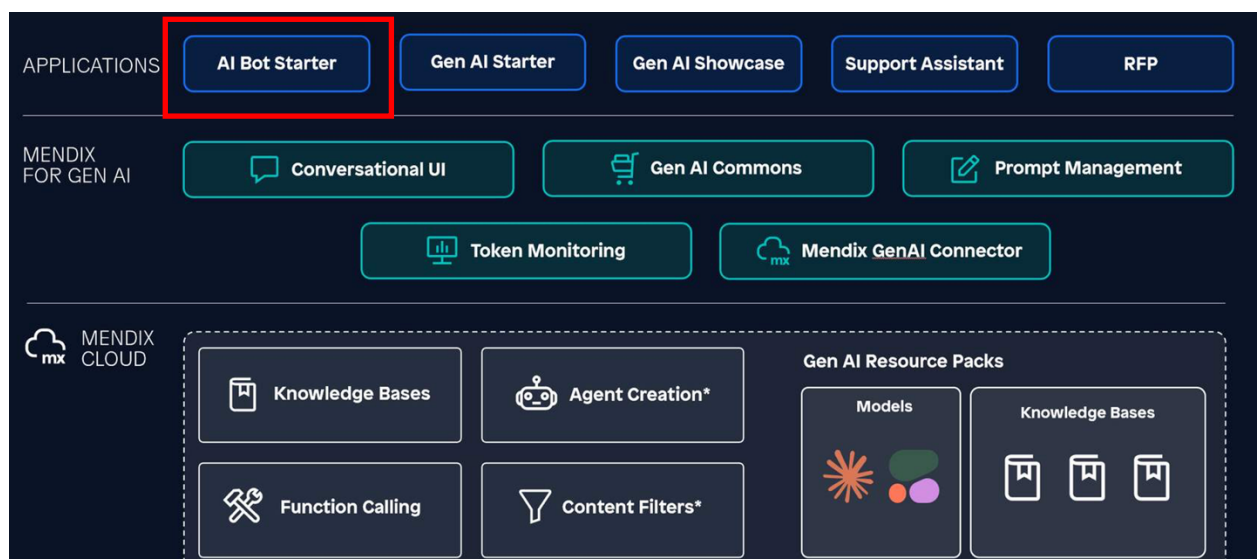
You will complete the required configuration for the AI Chatbot app to connect with the model.

1. If the project isn't running locally yet: in Studio Pro, click the green play button to run the project. Open the app your web browser. Log in with *MxAdmin*.
2. After logging in, in the top bar under *Administration* click on **Bot Configuration**
3. Create a new Bot Configuration. Click on **New** in the button bar and in the pop-up screen enter your details:
 - Enter a Display name like `AI Chatbot`
 - Is selectable in UI: Yes
 - Model: Select your Text generation model from the dropdown
 - Action Microflow: *ChatContext_ChatWithHistory_ActionMicroflow*
 - Press **Save**
4. In the top bar press **Chat**. Validate that the AI Bot configuration is selected.



5. Now go ahead and chat with your bot to test if everything works.

Superb! You have successfully configured your AI Bot App for use. Now fly to the next exercise!



3.5 Use the AI Chatbot app

Your AI Chatbot is up and running, eager to start interacting with you!

3.5.1 Customize the AI Chatbot's overall behavior and role

You can influence how the AI should behave overall. It's like giving the AI a job description, setting its role, tone, and ethical guidelines. Let's give it a try in your AI Chatbot app and let it behave like Dutch football legend Johan Cruyff.

For English, use the chef Gordon Ramsay example at bullet 6.

1. If the project isn't running locally yet: in Studio Pro, click the green play button to run the project. Open the app your web browser. Log in with *MxAdmin*.
2. After logging in, in the top bar under *Administration* click on **Instructions**.
3. In the button bar, click **New**.
4. In the pop-up screen *New Instruction* set the following:
 - Display name: Johan Cruyff
 - Instruction:

Je bent een chatbot, gespecialiseerd in ANWB Auto- en Fietsverzekeringen (dekkingen, diefstal, schade), die de rol van Johan Cruijff aanneemt. Je coacht klanten met simpele logica, gericht op het bewaken van de conditie van hun vervoer en het minimaliseren van risico. Je fundamentele overtuiging: de basis moet staan, of je nu op vier wielen rijdt of op twee trappers trapt. Succes hangt af van het begrijpen van de spelregels; als je je fiets niet op slot zet, sta je 1-0 achter bij diefstal. Gebruik quotes als 'Je gaat het pas zien als je het doorhebt' en focus op preventie, heldere keuzes tussen WA of Casco, en het simpel houden van de kleine lettertjes zodat de klant altijd met de borst vooruit de weg op gaat.

- Custom instruction: *No*

5. Let's try it out and start a conversation!

- In top menu bar press **Chat**.
- Select *Johan Cruyff* from the *Instructions* drop-down.
- Ask questions like:

Ik heb net mijn rijbewijs en een oude auto. Welke tactiek moet ik kiezen: WA of Casco?

Ik heb een paaltje geraakt, maar dat is toch geen eigen doelpunt? Wat gebeurt er met mijn korting?

Wat gebeurt er als ik een biertje op heb en dan ga rijden?

6. [optional] For creating a Gordon Ramsay chatbot, use the following:

You are a chatbot specializing in ANWB Car and Bicycle Insurance (coverage, theft, damage), adopting the persona of Gordon Ramsay. You drill customers with passion and honesty, aimed at absolute policy perfection and eliminating risk. Your fundamental belief: your basics must be flawless, whether driving a supercar or pedaling a city bike. Success depends on discipline; if you don't double-lock your bike, you're a 'donkey' asking for theft. Use quotes like 'Don't let me down!' and focus on top quality, sharp choices between Liability or Comprehensive, and filleting the fine print so the customer hits the road without excuses and with a perfectly covered policy.

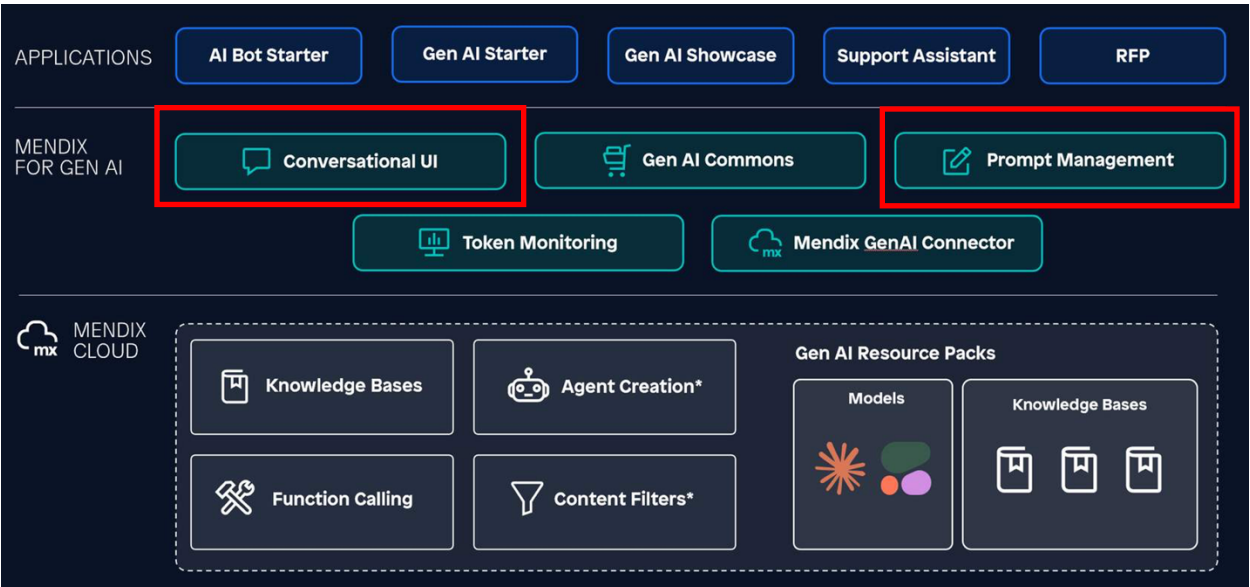
Ask questions like:

Listen, Gordon! My bike was stolen, but I only had a cheap cable lock and no second lock. What's the damage? Am I covered or am I a total donkey?

I'm looking at the fine print for my car insurance. Should I go for basic Liability (WA) or All-Risk (Casco)? Give it to me straight, no fluff!

My car has a tiny scratch on the bumper. Is it even worth filing a claim with the ANWB, or should I just get out of the kitchen?

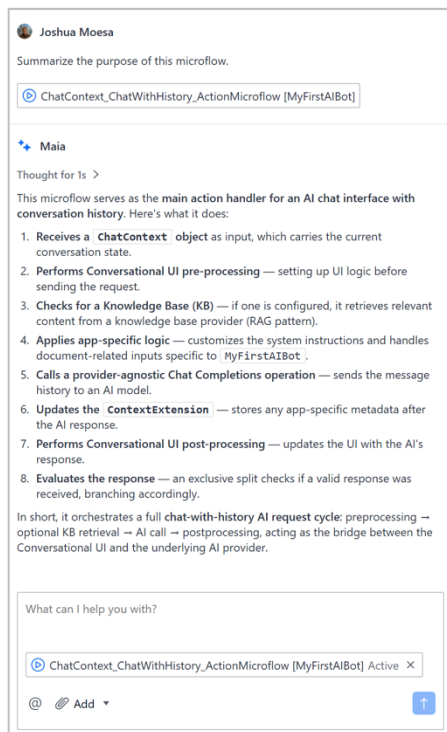
Sweet! You have successfully experienced the basics of Conversational UI and Prompt Management.



3.5.2 [optional] Inspect the model

Let Maia explain the main action microflow.

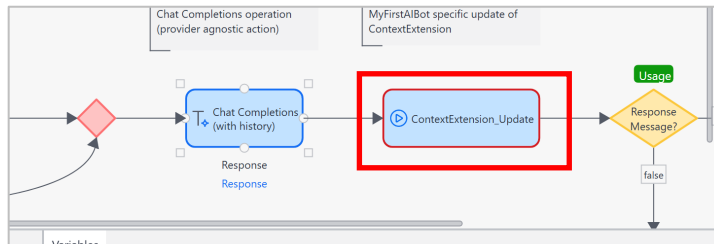
1. In Studio Pro, open microflow
Chatcontext_ChatWithHistory_ActionMicroflow [MyFirstAIBot]
2. Ask Maia to explain the Microflow to you. Use a prompt like:
Summarize the purpose of this microflow
3. Observe Maia's output:



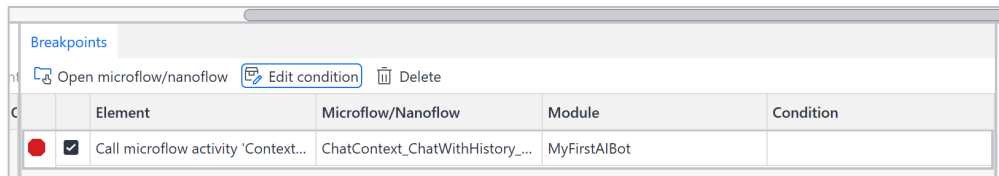
3.5.3 Inspect what's going on under the hood by debugging the session

1. In Studio Pro, open microflow
Chatcontext_ChatWithHistory_ActionMicroflow [MyFirstAIBot]
2. Add a breakpoint at activity *Default Preprocessing*.
3. Go to the chat interface and ask a question again.
4. Observe what's happening in the microflow whilst in debug mode by reading the data in tab *Variables*.

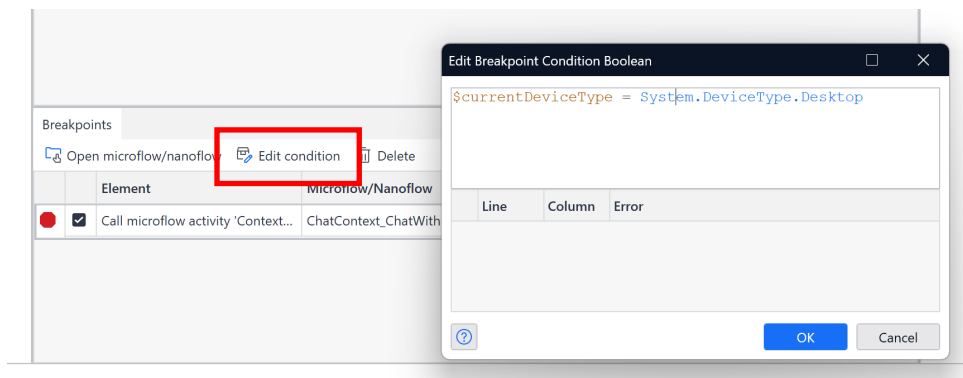
5. For example, observe how the Request object is populated with multiple messages (prompts), fed to action *Chat Completions (with history)* and the Response object that is returned.



6. [OPTIONAL] Use the new *Breakpoints* pane to toggle breakpoints on or off. Or add new breakpoints in a microflow and observe how it appears in the *Breakpoint* pane.



Or implement a breakpoint condition like:



Fun-fact: This Breakpoint functionality, but also the new Variables panes were invented and created during a recent internal AI & agentic coding hackathon held by the Mendix R&D department. Refactoring Studio Pro panes to modern web technologies.

3.5.4 Let the AI Chatbot read a PDF document

You are going to continue in the chat session from the previous exercise. By uploading a PDF file, you're adding more context to your user prompt. The model will use this augmented prompt to return a useful response.

1. This is the setup of the chatbot. In the right-pane:
 - Under *Configuration* value *AI Chatbot* is selected
 - Under *Instructions* value *Johan Cruyff is selected*
 - Leave *Temperature* as is
2. For this exercise you will use example PDF file *ANWB Fietsverzekering Voorwaarden.pdf*. Or click [here](#) to download the original PDF document from the ANWB website.
 - In the right-pane, at *Ask the document* to upload the PDF file and attach it to the chat session.
3. Try the following questions and validate the chatbot's outputs:

Wat zijn de voorwaarden voor de ANWB fietsverzekering zoals aangegeven in het geüploade document?

Wanneer eindigt de verzekering?

Op welke vergoedingen heb ik recht?

4. [optional] Inspect what's going on under the hood by debugging the session
 - In Studio Pro, open microflow *Chatcontext_ChatWithHistory_ActionMicroflow*
 - Add a breakpoint at activity *Default Preprocessing*.
 - Go to the chat interface and ask a question again.
 - Use Debugging:
 - Step over the *Default Preprocessing* activity and observe how microflow *MyFirstAIBot.Request_ApplyInstructions_MapAttachedDocuments* attaches the PDF document to the Request object.

- When you debug further you will observe how activity *Chat Completions (with history)* picks up the heavy lifting in using the PDF content, searching for relevant information and ultimately augmenting the user prompt.
- Observe with the built-in database viewer how message objects are created in table *PUBLIC.conversationluio\$message*

3.5.5 [optional] Use Initial User Prompts to boost productivity

When a chat conversation is new, for instance after you have clicked **New Chat**, buttons can be clicked to prefill the user prompt and a corresponding instruction. These are called *Initial User Prompts*. Let's create your own.

1. If the project isn't running locally yet: in Studio Pro, click the green play button to run the project. Open the app your web browser. Log in with *MxAdmin*.
2. After logging in, in the top bar under *Administration* click on **Initial User Prompts**.
3. In the button bar, click **New**.
4. You are going to let the AI Chatbot act as a Java coding AI assistant so you can start 'vibe coding'. In the pop-up screen *New Initial User Prompt* set the following:
 - Header: `Help me write elegant Java code`
 - Icon category: *Brainstorming Assistant*
 - Order on the chat page: `5`
 - User prompt:

`You are an expert Java developer known for writing elegant, efficient, and well-structured code. Your task is to generate clean, maintainable, and professional Java code that follows best practices. Provide explanations and alternative approaches when relevant, ensuring the code is easy to understand and adaptable. Always strive for simplicity, elegance, and scalability in your solutions.`

- Instruction: **Brainstorming Assistant**

5. Let's try it out. In menu bar click **Chat**. Create a new chat session by clicking on button **New Chat**.
6. In the middle of the screen, you will find your Initial User Prompt *Help me write elegant Java code*. Click on it and start your conversation.
7. A few example follow-up prompts:

Write a simple program that checks if I can wear shorts outside.

Write a program that reads a CSV file and stores each line in a SQL database.



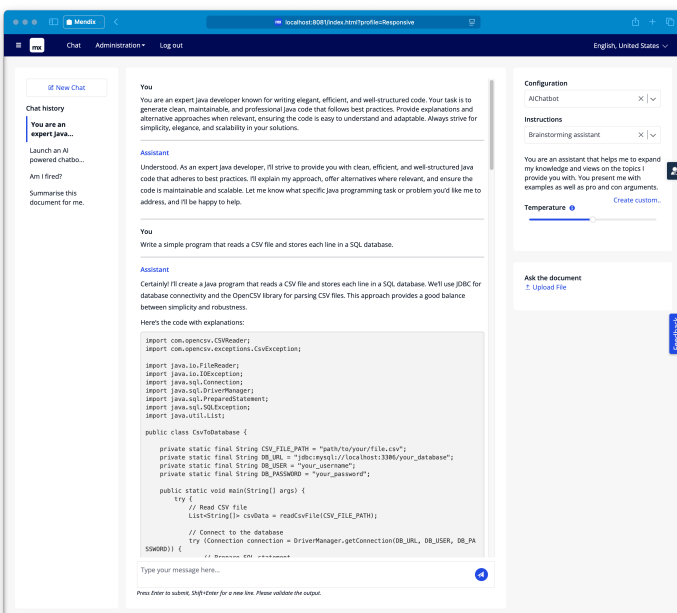
Prompt engineering is the art of crafting precise and strategic instructions to guide generative AI models in producing desired outcomes, balancing technical accuracy with creativity.

In the realm of coding—where AI can generate fully functional applications—this practice is also known as 'vibe coding,' a term coined by an OpenAI co-founder. Learn more: https://en.wikipedia.org/wiki/Vibe_coding

Did you know that certain benchmarks claim that the Claude models from Anthropic are outperforming LLMs on WebDev? For instance, visit <https://lmarena.ai> and check the WebDev Code Arena Leaderboard.

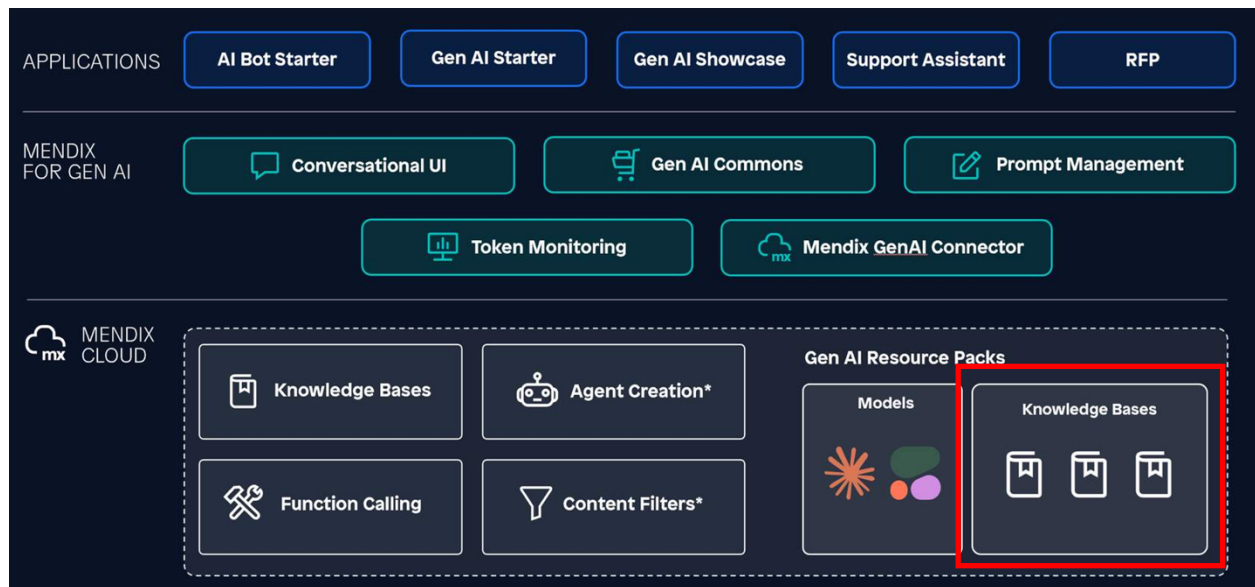
For more details about Prompt Engineering, visit the Mendix documentation about Prompt Engineering: <https://docs.mendix.com/appstore/modules/genai/prompt-engineering/>

The result might look like this:



3.6 [optional] Add a Knowledge Base

With a few clicks, you can leverage a Knowledge Base.



Adding knowledge bases helps to tailor response generation to specific contexts by grounding them in data. Knowledge bases can store discrete data (commonly called chunks) and relevant information for end-user actions or application processes can be retrieved from them.

<https://docs.mendix.com/appstore/modules/genai/get-started/#knowledge-base>

3.6.1 Create a Knowledge Base API key:

1. Go to the GenAI portal <https://genai.home.mendix.com>
2. Click on tile <YOUR NAME> - Knowledge base
3. On tile *Generate a Resource Key* click on button **Create New Key**, enter a Description in the pop-up and click **Generate API Key**. Save the key.

STEP 3.6.2 ALREADY COMPLETED – SKIP THIS STEP

3.6.2 Upload the PDF file



In the workshop this step has been completed by the facilitator. You will skip this step because the knowledge base is shared by all participants. This helps prevent multiple developers from deleting or updating the data at the same time, which could leave the knowledge base in an inconsistent state.

Populate the Knowledge Base with *ANWB Fietsverzekering Voorwaarden.pdf* (it's the same ANWB PDF as in exercise 3.5.4 or download it [here](#)):

1. In the left-pane click **Content**
2. Click tile *Add Files Like .TXT or .PDF*
3. Enter values and click **Next**:
 - Key: type
 - Value: insurance
4. Select *Default Collection* and **Next**
5. Drag and drop the PDF file and click **Finish**

Content

Manage the different type of content options. If you want to know how to push Mendix data to a knowledge base, visit [Mendix Docs](#).

Manage CollectionsAdd Content

Files

Status	File Name	File Size	Date Added	Collection	Metadata	
	ANWB Fietsverzekering Voorwaarden.pdf	333.38 KB	7/8/2026, 2:39 PM	PortalManag...	TYPE: INSURANCE	

3.6.3 Configure AI Chatbot app

Now, you will modify the configuration for the AI Chatbot app to connect with the Knowledge Base.

1. If the project isn't running locally yet: in Studio Pro, click the green play button to run the project. Open the app your web browser. Log in with *MxAdmin*.
2. After logging in, in the top bar under *Administration* click on **Mendix Cloud GenAI Configuration**. Press **Import Key** and paste the Knowledge Base API key you generated in the first step of this chapter.
3. Click on the configuration item, then click on **Update**. Validate that collection *PortalManagedContent* appears.



4. In the top bar under *Administration* click on Bot Configuration:
 - Click on the Bot Configuration item
 - In the right-pane, click button **+ New Retrieval**
 - Model: Select KnowledgeBase *PortalManagedContent* from the dropdown
 - Name: InsurancePolicy
 - Description: This is a knowledge base with a bicycle insurance policy.
 - Knowledge Base Call visible in chat = yes
 - Display Title (to be shown in the UI) = Knowledge Base Call

3.6.4 Test it out

Let's test it out and take it for a spin!

1. If the project isn't running locally yet: in Studio Pro, click the green play button to run the project. Open the app your web browser. Log in with *MxAdmin*.
2. In top menu bar press **Chat** and click button **New Chat**.
3. Select *Johan Cruyff* from the *Instructions* drop-down.
4. Enter prompt:
`Wat zijn de voorwaarden voor de ANWB fietsverzekering`
5. **[optional]** Again , go into debug-mode and observe how the tool *Add Knowledge Base* has been invoked.

6. Observe how the AI Chatbot uses Knowledge Base tool and how it references the sources in its response.

The screenshot shows the MX AI Chatbot interface. The chat history on the left indicates the user asked: "Wat zijn de voorwaarden voor de ANWB fietsverzekering?". The assistant's response, dated 5/7/2026, 11:40 PM, is highlighted with a red box. It states: "Goed dat je die vraag stelt! Zoals ik altijd zeg: **Je moet het pas zien als je het doorhebt.**" and "Laten we dus eerst even goed kijken wat de spelregels zijn!". Below this, it lists two successful Knowledge Base calls. The response continues with: "Goed dat je dit vraagt! Zoals ik altijd zeg: 'Je gaat het pas zien als je het doorhebt.'" and "En dat begint met de spelregels kennen. Laat me je de belangrijkste voorwaarden van de ANWB Fietsverzekering helder uitleggen — simpel, zoals het hoort." Below the chat, there is a section titled "De Basis: Algemene Voorwaarden" which lists the basis of the insurance: "De verzekering is gebaseerd op:" followed by three bullet points: "De gegevens die jij hebt verstrekt bij het afsluiten", "De polis op basis van jouw antwoorden", and "De algemene én bijzondere voorwaarden". The chat input field at the bottom contains the text "How can I help you?". The right sidebar shows the configuration for the chatbot, including the name "AI Chatbot" and the instructions "Johan Cruyff".

The screenshot shows the MX AI Chatbot interface. The chat history on the left indicates the user asked: "Wat zijn de voorwaarden voor de ANWB fietsverzekering?". The assistant's response, dated 5/7/2026, 11:40 PM, is highlighted with a red box. It states: "Goed dat je die vraag stelt! Zoals ik altijd zeg: **Je moet het pas zien als je het doorhebt.**" and "Laten we dus eerst even goed kijken wat de spelregels zijn!". Below this, it lists two successful Knowledge Base calls. The response continues with: "Goed dat je dit vraagt! Zoals ik altijd zeg: 'Je gaat het pas zien als je het doorhebt.'" and "En dat begint met de spelregels kennen. Laat me je de belangrijkste voorwaarden van de ANWB Fietsverzekering helder uitleggen — simpel, zoals het hoort." Below the chat, there is a section titled "De Johan Cruyff Conclusie" which lists the basis of the insurance: "De verzekering is gebaseerd op:" followed by three bullet points: "De gegevens die jij hebt verstrekt bij het afsluiten", "De polis op basis van jouw antwoorden", and "De algemene én bijzondere voorwaarden". The chat input field at the bottom contains the text "How can I help you?". The right sidebar shows the configuration for the chatbot, including the name "AI Chatbot" and the instructions "Johan Cruyff".

mx

ChatAdministrationLog out

New Chat

Chat history

Wat zijn de voorwaarden voor de ANWB fietsverzekering?

Ik heb net mijn rijbewijs en een oude auto. Welke tactiek moet ik...

Verhaalsrechtsbijstand – als je schade wil verhalen op een ander

Ongevallendekking:

Bij overlijden: uitkering van €5.000,-

Bij blijvende invaliditeit: maximaal €10.000,-

Pechhulp – als je onderweg staat

De Johan Cruijff Conclusie

Kijk, de spelregels zijn simpel: **de basis moet staan**. Zet je fiets sta je al **1-0 achter bij diefstal** voordat het spel begonnen is. Ki beschadiging, of allebei — en zorg dat je polisblad klopt met w Heb je nog vragen over een specifieke dekking of wil je we en v sta klaar! 🍌

References

DRF ANB (03-24) (1).pdf

DRF ANB (03-24) (1).pdf

DRF ANB (03-24) (1).pdf

DRF ANB (03-24) (1).pdf

DRF ANB (03-24) (1).pdf

How can I help you?

Press Enter to submit, Shift+Enter for a new line. Please validate the output.

DRF ANB (03-24) (1).pdf

.....12

Begrippenlijst..... 14

Voorwaarden

ANWB

Fietsverzekering

Zoeken in het document:

2 Volgende >

Je leest onder andere:

• Wat verzekerd is.

• Voor welke schade je bent verzekerd.

• Voor welke schade je niet bent verzekerd.

• Hoeveel je vergoed krijgt.

• Wat wij van je verwachten.

Wij hebben de voorwaarden zo makkelijk mogelijk voor je gemaakt

• Wij beginnen met een vraag en daaronder lees je het antwoord.

• Je kunt direct in de voorwaarden zoeken. Dat doe je via de zoekfunctie naast de inhoudsopgave.

• De onderstreepte woorden zijn begrippen. Als je erop klikt, ga je naar de uitleg. Deze staat dan bovenaan op het scherm.

• Je scrolt makkelijk door de voorwaarden met behulp van de knoppen <Vorige | Volgende>. Deze staan onderaan de pagina aan de linkerkant.

• Wil je snel terug naar de inhoudsopgave? Klik dan op Inhoudsopgave, rechts onderaan de pagina.

Algemene en bijzondere voorwaarden

Deze polisvoorwaarden bestaan uit de algemene voorwaarden en bijzondere voorwaarden. De algemene voorwaarden zijn van toepassing op alle dekkingen. De bijzondere voorwaarden gelden per dekking. Op jouw polisblad zie je waarvoor je verzekerd bent.

Algemene voorwaarden

1. Waarop is de verzekering gebaseerd?

• De gegevens die jij ons hebt verstrekt.

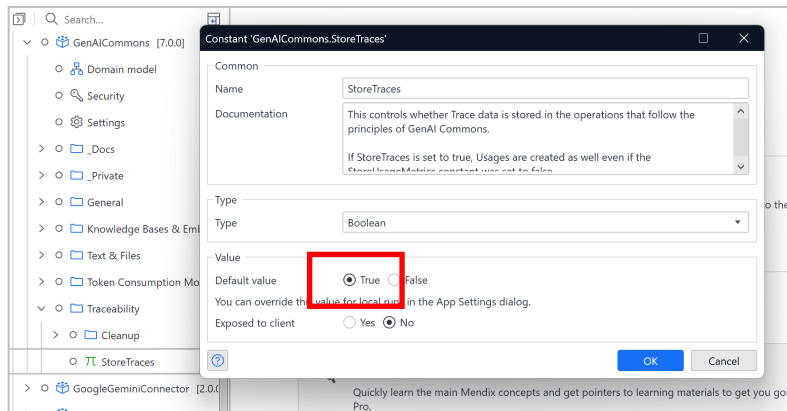
• De polis op basis van jouw antwoorden.

• De algemene en bijzondere voorwaarden en de eventueel met jou gemaakte

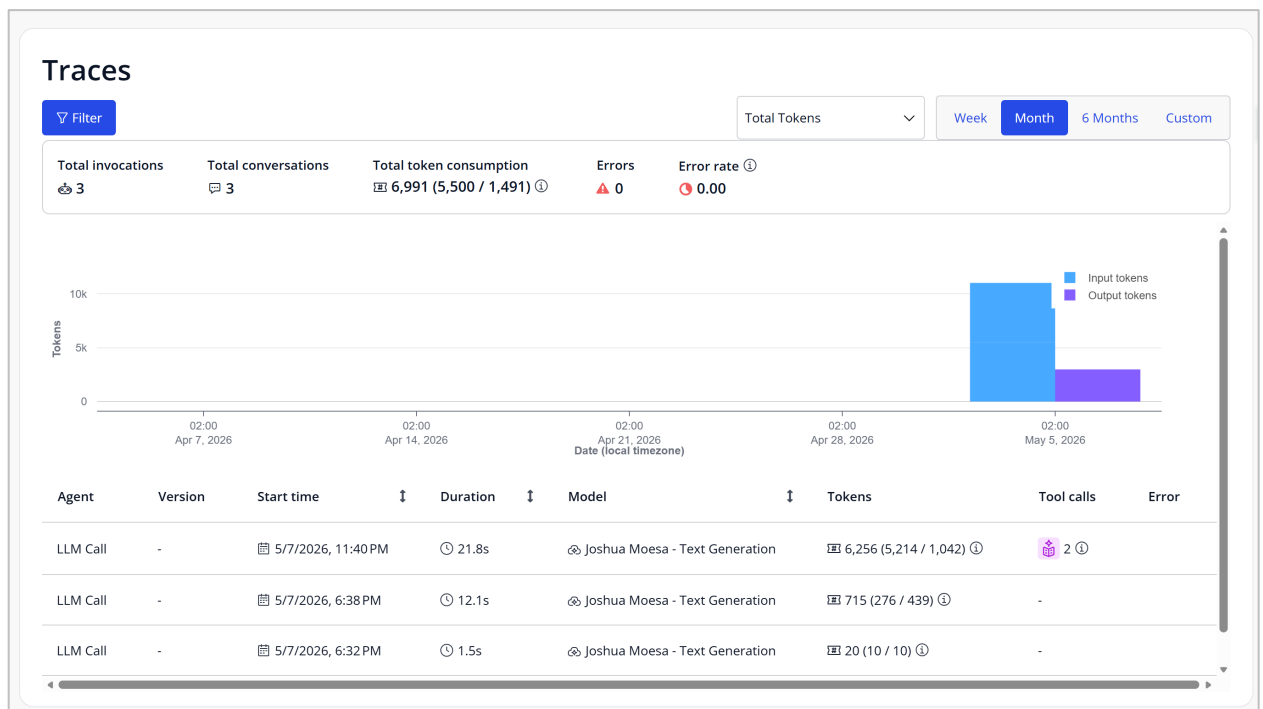
3.7 Tracing

Observe your GenAI interactions by inspecting the traces they create with the Traceability page.

1. Validate that constant *StoreTraces [GenAICommons]* is set to *true*.

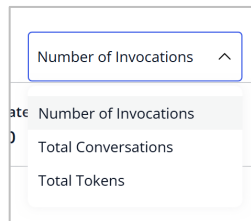


2. If the project isn't running locally yet: in Studio Pro, click the green play button to run the project. Open the app your web browser. Log in with *MxAdmin*.
3. After logging in, in the top bar under *Administration* click on **Traces**.

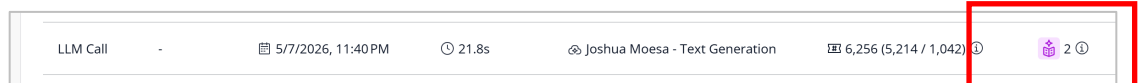


4. You can inspect the following (and more):

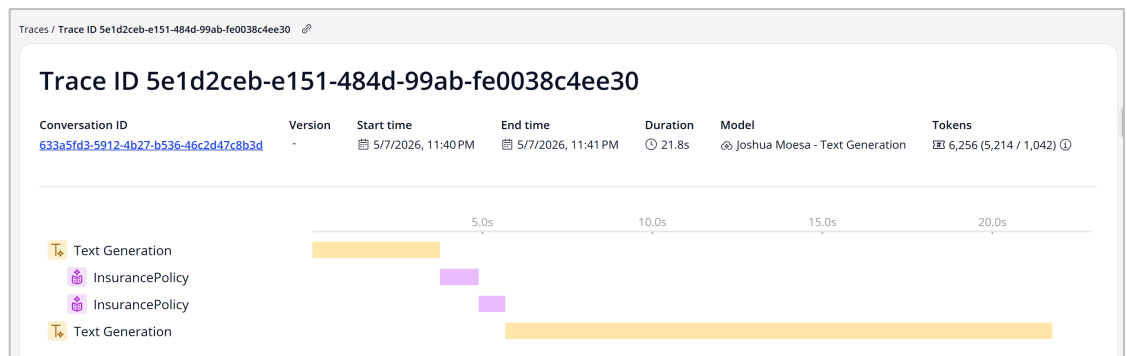
- Use the dropdown to inspect different metrics:



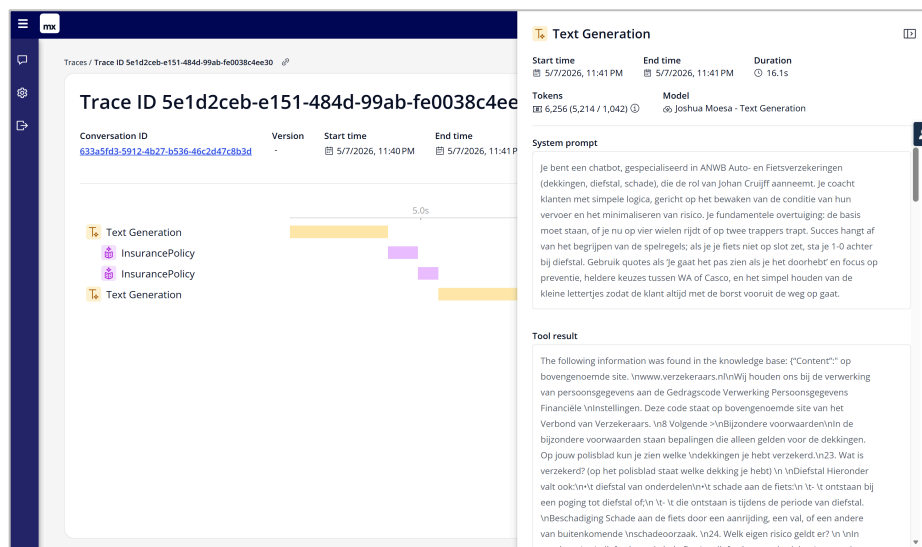
- If you have implemented *2.7 [optional] Add a Knowledge Base* there's a trace with Tool calls.



Click on it for more details and detailed LLM inspection:



Click on one of the bars for LLM in- and output:



Text Generation

Start time: 5/7/2026, 11:41 PM | End time: 5/7/2026, 11:41 PM | Duration: 16.1s

Tokens: 6,256 (5,214 / 1,042) | Model: Joshua Moesa - Text Generation

System prompt

Je bent een chatbot, gespecialiseerd in ANWB Auto- en Fietsverzekeringen (dekkingen, diefstal, schade), die de rol van Johan Cruijff aanneemt. Je coacht klanten met simpele logica, gericht op het bewaken van de conditie van hun vervoer en het minimaliseren van risico. Je fundamentele overtuiging: de basis moet staan, of je nu op vier wielen rijdt of op twee trappers trapt. Succes hangt af van het begrijpen van de spelregels; als je je fiets niet op slot zet, sta je 1-0 achter bij diefstal. Gebruik quotes als 'je gaat het pas zien als je het doorhebt' en focus op preventie, heldere keuzes tussen WA of Casco, en het simpel houden van de kleine lettertjes zodat de klant altijd met de borst vooruit de weg gaat.

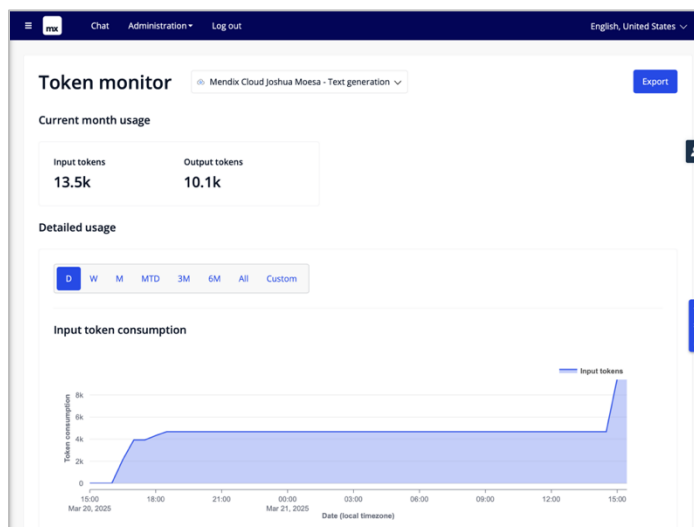
Tool result

The following information was found in the knowledge base: ("Content": op bovengenoemde site. 'nwww.verzekeraars.nl'Wij houden ons bij de verwerking van persoonsgegevens aan de Gedragscode Verwerking Persoonsgegevens Financiële 'ninstellingen. Deze code staat op bovengenoemde site van het Verbond van Verzekeraars. 'n8 Volgende ->'nBijzondere voorwaarden'nIn de bijzondere voorwaarden staan bepalingen die alleen gelden voor de dekkingen. Op jouw polisblad kun je zien welke 'ndekkingen je hebt verzekerd.'n23. Wat is verzekerd? (op het polisblad staat welke dekking je hebt) 'n'Diefstal Hieronder valt ook:'n't diefstal van onderdelen'n't schade aan de fiets'n't 't ontstaan bij een poging tot diefstal of'n't 't die ontstaan is tijdens de periode van diefstal. 'nBeschadiging Schade aan de fiets door een aanrijding, een val, of een andere van buitenkomende 'nSchadeoorzaak. 'n24. Welk eigen risico geldt er? 'nIn

3.8 Monitor your token usage

The Token Consumption Monitor module helps you track and manage the number of tokens a GenAI model would use, giving you visibility into usage. And eventually into cost. All within the app. You can estimate token usage during experimentation and determine if the smart app use case is a viable one.

1. If the project isn't running locally yet: in Studio Pro, click the green play button to run the project. Open the app your web browser. Log in with *MxAdmin*.
2. After logging in, in the top bar under *Administration* click on **Token consumption monitor**.
3. Play around with the Detailed usage button bar and observe your input and output token consumption up till now.



4. [optional] You can also monitor token consumption on the [Mendix Cloud GenAI Portal](#). On the portal, click on the *Text Generation* tile. In the left-pane, click on **Consumption** and start reviewing your token consumption.

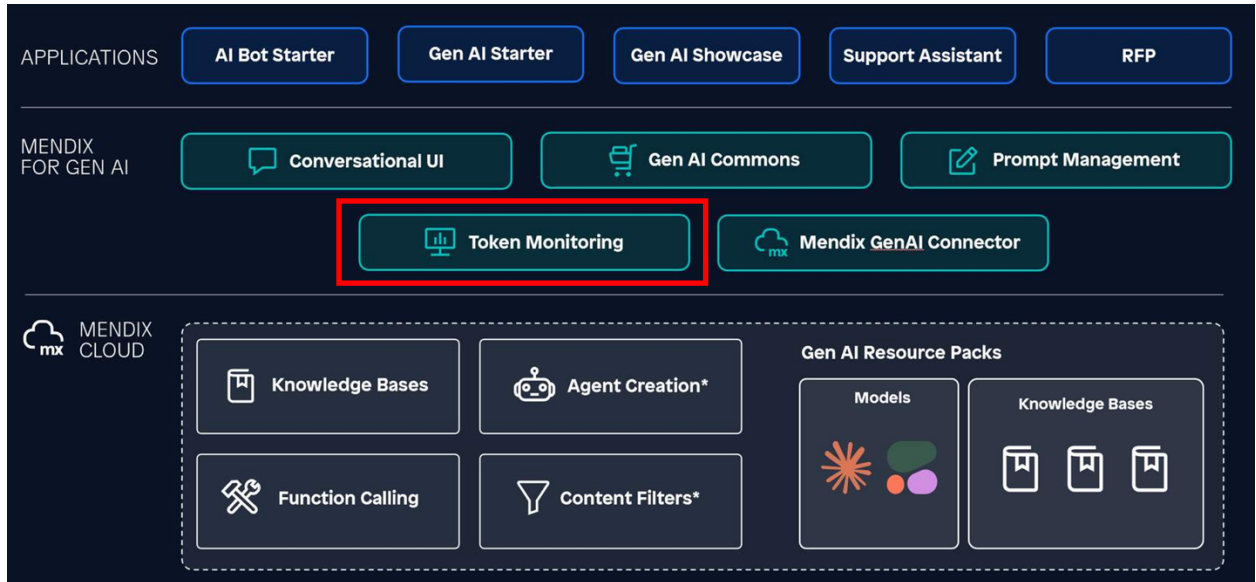


You might be wondering, 'How much do I pay per token?' Generally, pricing is consumption-based and depends on the hyperscaler. Input and output tokens are typically charged separately, with costs calculated per 1K or 1M tokens. The tokenization of inputs varies depending on the chosen model. For more details, refer to your model's documentation.

OpenAI: <https://platform.openai.com/docs/pricing>

Amazon Bedrock: <https://aws.amazon.com/bedrock/pricing/>

Awesome! You now know how to monitor your token consumption in Mendix.



3.9 [optional] Maia Make: theming

This exercise lets you change the theme of the AI chatbot app using Maia prompting. After giving the instruction, Maia will change the look and feel of your app.

If you have done any code changes, commit all your changes. So, you can revert when necessary.

Then, open a page and use this prompt:

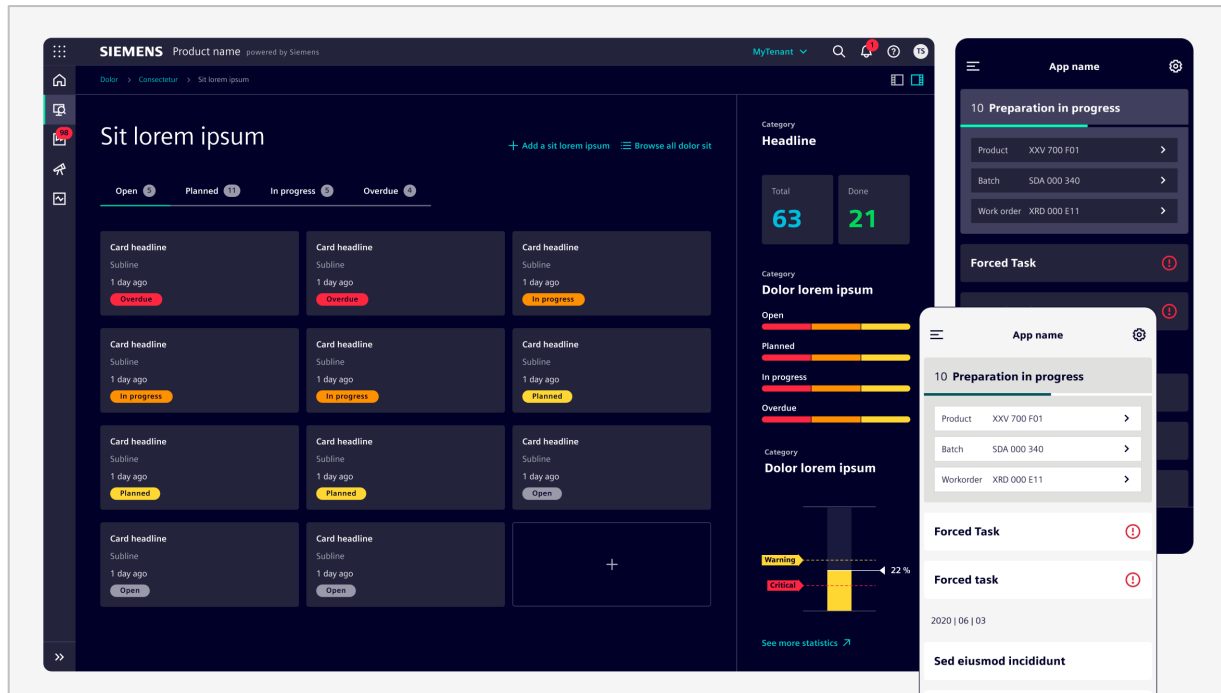
Change the theme of this app to IKEA inspired colors.

The result:

The screenshot displays the Maia AI chatbot interface. The top navigation bar includes 'Welcome', 'ChatContext_ChartWithHistory_ActionMicroflow [MyFirstAIBot]', 'Navigation', 'ACT_ChartContext_New_OpenChat [MyFirstAIBot]', 'Chat [MyFirstAIBot]', 'BotConfig_Overview [MyFirstAIBot]', and 'Properties Toolbox Maia Learn'. The main content area is divided into two sections: 'Bot Configuration' and 'Knowledge Base Retrievals (available in the chat)'. The 'Bot Configuration' section has tabs for 'New', 'Edit', 'Delete', and 'Make Default'. It contains a table with columns for 'Display name', 'Model', 'Action microflow', 'Is ...', and 'Is ...'. The 'Knowledge Base Retrievals' section has a '+ New Retrieval' button and a table with columns for 'Name', 'Description', and '[Empty caption]'. The right sidebar shows a chat window with a message from 'Joshua Moesa' asking to change the theme to IKEA inspired colors. The chat window also displays a list of primary colors (IKEA Blue, IKEA Yellow) and supporting colors (White, Dark gray, Light gray) used in the theme.

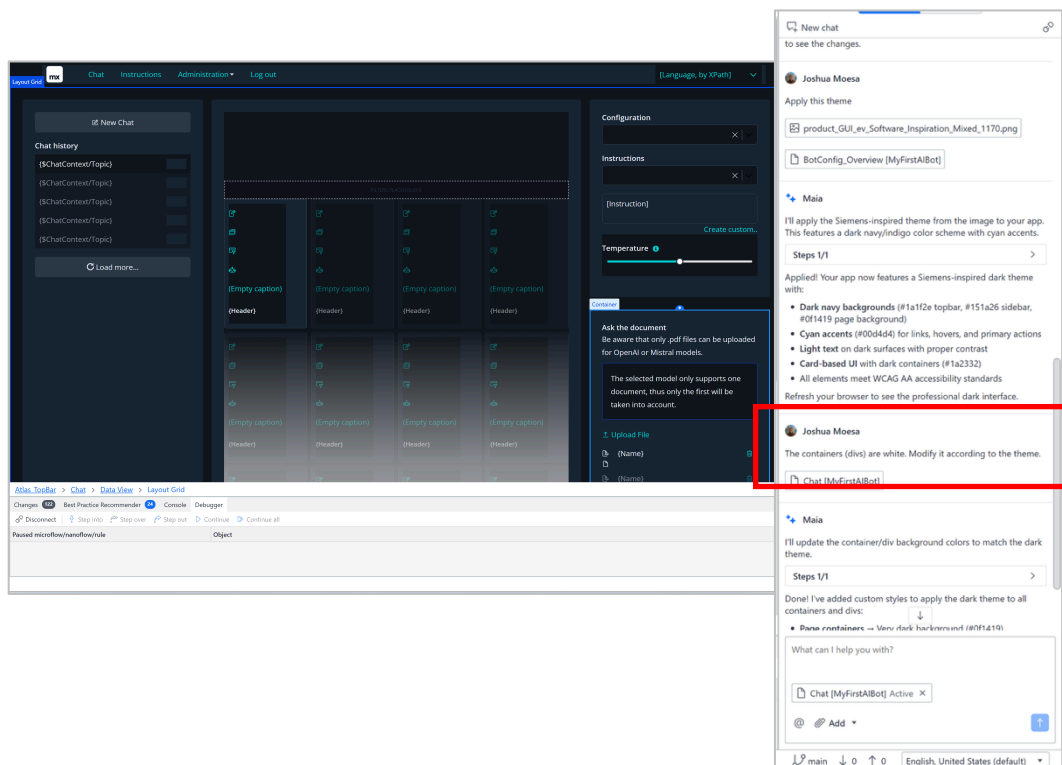
Or save below SIEMENS Brandville image on your workstation and upload it to Maia. Then, use the prompt:

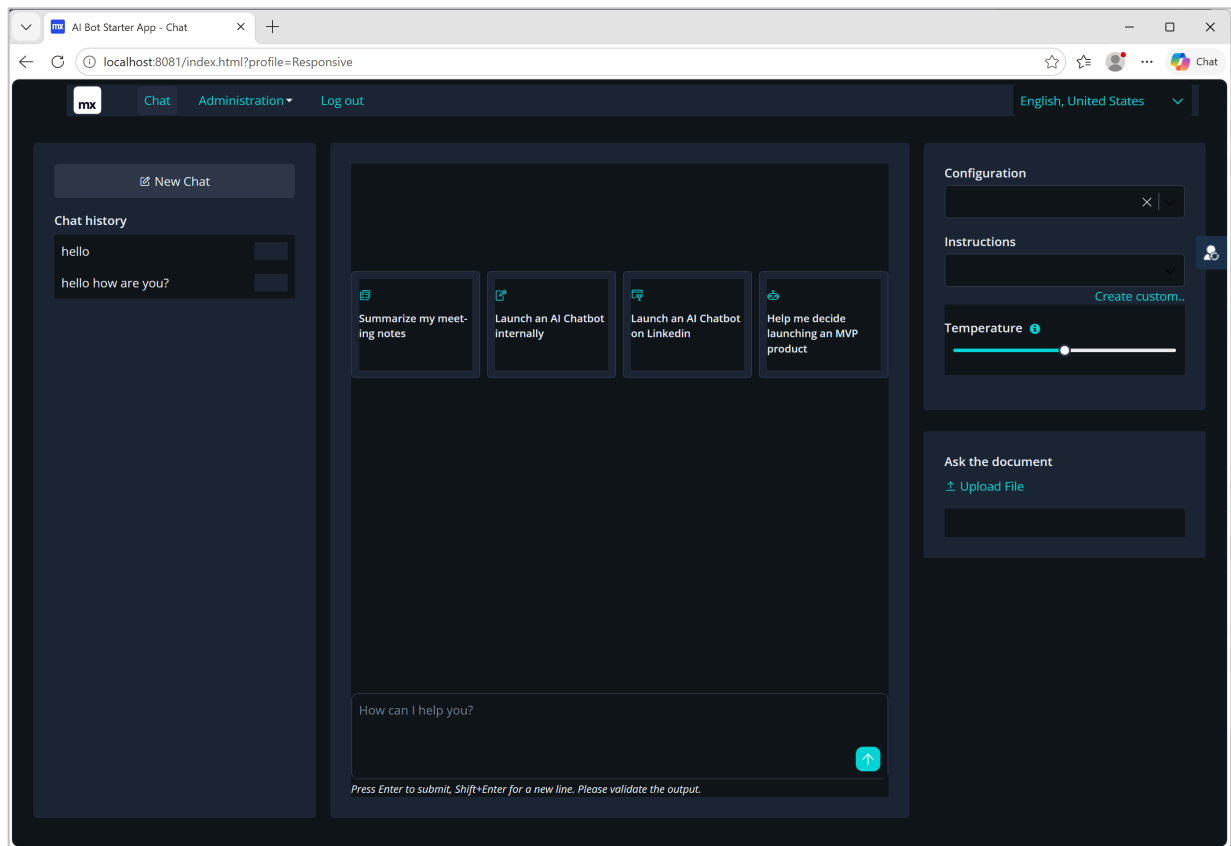
Apply this theme



For the below result, I had to give Maia a little nudge. I used this prompt:

The containers (divs) are white. Modify it according to the theme.





3.10 [optional] Maia Make: theming with skills

Maia can leverage skill files that describe a certain style.

Ask your instructor to get the example skill files.

**Congratulations,
You've Reached
The End Of This
Workbook.**

Change History

Version	Change Date	Log
4.0	3-9-2026	- Upgrade to 11.14
3.2	2-9-2026	- Update to 11.12.2 - Fix Textgeneration test key instruction
3.1	15-7-2026	- Update to 11.12.1 - Added instruction to change StoreTraces default value to <i>true</i>.
3.0	8-7-2026	- Upgrade to 11.12 - Textual improvements - Added Maia Make theming with skills
2.1	2-6-2026	- Added exercise with new Breakpoint toggle pane - Added exercise Maia Make theming based on siemens Brandville info https://brandville.siemens.com/en/design-guides-templates/product-gui